Stagnone di Marsala digital twin: project resumption and status

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To: Prof. Giuseppe Ciraolo, Prof. Mauro De Marchis 7 August 2026

1. Summary

Work on the coupled three-dimensional model resumed on 5 August after a pause since mid-June. The main outcome of the past three days is that the first manuscript now has a complete draft, in LaTeX, formatted for *Estuarine, Coastal and Shelf Science* and within all of that journal's limits. One methodological question remains open and is under test as this is written.

The manuscript is provisionally titled **"Transport validation of a wave-coupled lagoon model: interacting effects of seagrass roughness and bed mobility"**, with the three of us as authors. I would be grateful for your comments on both the framing and the content.

2. What the study asks and what it found

Coastal lagoon models are routinely validated against water level and then used to describe transport. The paper asks whether the first licenses the second. Six model configurations were run over the same nine days of July 2025, from the same initial condition and forcing, differing only in wave coupling, bed mobility, and whether seagrass resistance is uniform or distributed from a satellite classification. Each was scored against three tide gauges and against 35 GPS surface drifters released over twelve deployments.

Water level does not separate the configurations. Anomaly RMSE differs by six millimetres or less between members at any station, which is within the measurement uncertainty, and no configuration is consistently best across the three stations. Over the same simulations Liu–Weisberg skill spans 0.505 to 0.655 and endpoint separation spans 334 to 661 m. A calibration conducted on water level alone would have had no basis on which to prefer any of them.

No treatment improves Lagrangian skill on its own. Distributed seagrass roughness on a fixed bed changes skill by 0.009 or less, bed mobility under uniform roughness by 0.019, and wave coupling reduces it by 0.035 to 0.042. Applied together, distributed roughness and bed mobility raise skill by 0.131 to 0.149, improving 32 of the 35 drifters, and only that combination reproduces the observed transport rate.

Splitting the trajectory error into magnitude and direction shows what each treatment corrects, which is the most recent result and the one I consider the paper's contribution.

Configuration	Path length (sim/obs)	Net displacement (sim/obs)	Heading error (deg)
Four fixed-bed members	0.77–0.80	0.78–0.82	15.8–16.2
Mobile bed, uniform roughness	0.94	0.95	12.5
Mobile bed, distributed roughness	0.98	0.99	**8.4**

The four fixed-bed members coincide in all three measures. Bed mobility acts on magnitude, restoring the distance travelled. Distributed roughness acts on direction, and does so only once the bed can move: on a fixed bed it leaves the heading error unchanged to the first decimal. This is reported as a narrowing rather than a complete mechanism, since reconstructing endpoint separation from a mean displacement ratio and a mean heading still underpredicts the two mobile-bed members by a factor of 1.4 to 1.9.

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Three simpler explanations were tested and rejected: the spatial pattern of bed change is indistinguishable between the two mobile-bed members, canopy drag does not suppress erosion beneath the meadows, and the successful configuration does not carry a stronger mean current field.

Two further results support the modelling choices. The canopy parametrisation reduces mean wave orbital velocity at the bed by 14.3% over *Posidonia*, 13.7% over *Cymodocea*, 11.3% over sand and 3.3% over exposed rock, an ordering that follows canopy density. And surface current speed exceeds bed speed by a factor of 1.9 to 4.5 at sub-metre mean depth, consistent with De Marchis et al. (2012) in the same lagoon, which is the quantitative justification for solving in three dimensions.

3. Open item

The experimental design is a factorial in three factors, of which six of the eight cells were obtained. The two missing cells combine a mobile bed with no wave coupling. Both abort on the model's velocity cap, at 26.96 and 43.42 $\mathrm{m\,s^{-1}}$ against a limit of 25, and they do so deterministically: two independent attempts stop at the same simulated instant with velocities identical to six decimal places. The four wave-coupled members, carrying the same sediment settings, complete without approaching their cap.

Diagnosis so far places the instability offshore and diffuse rather than at a single defective cell. Of the 150 fastest cells before the abort, none is inside the lagoon and none is above the datum, and their median depth is 34 m. Two candidate causes are being tested at the moment, both concerning the offshore sector next to the open boundary: suspended sediment feeding the density field, and the treatment of the tangential velocity component at boundary faces. Results are expected within hours of writing.

If neither resolves it, the result is reportable as an observed bound on the design rather than a technical failure, and the manuscript already carries it that way. The consequence, stated explicitly in the text, is that the wave contribution is measured on a fixed bed only.

4. Manuscript status and what I would ask of you

The draft is complete across all six sections, with eight figures and three tables, 6200 words of body text against a limit of 8000, and 24 references verified against the source PDFs. Highlights and a cover letter are drafted. Outstanding are a graphical abstract, which *ECSS* requires, and the WETWISE project identifier for the funding statement.

I would welcome your reading on three points in particular:

1. Whether the framing holds. The paper claims a validation-strategy result, not a description of this lagoon, and the journal does not accept manuscripts of mainly local interest.
2. Whether reporting the bed-mobility result is wise given that its critical shear stress for erosion has not been constrained against a repeated bathymetric survey. The manuscript states this openly, and the lagoon bed changes by up to 1.0 m over nine days in isolated cells, which is not a plausible rate. Satellite-derived bathymetry over repeated epochs is the route I would propose to close it.
3. The CRediT roles I have assigned, which follow those of the reservoirs manuscripts.

The full draft, figures and analysis code are in the project repository, and I can send the compiled PDF on request.

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